



**ENT-164** · Intro to Making

**Tufts**  
UNIVERSITY

Derby Entrepreneurship  
Center at Tufts

CLASS 4 · WEEK 4 · FALL 2026

# 3D Printing

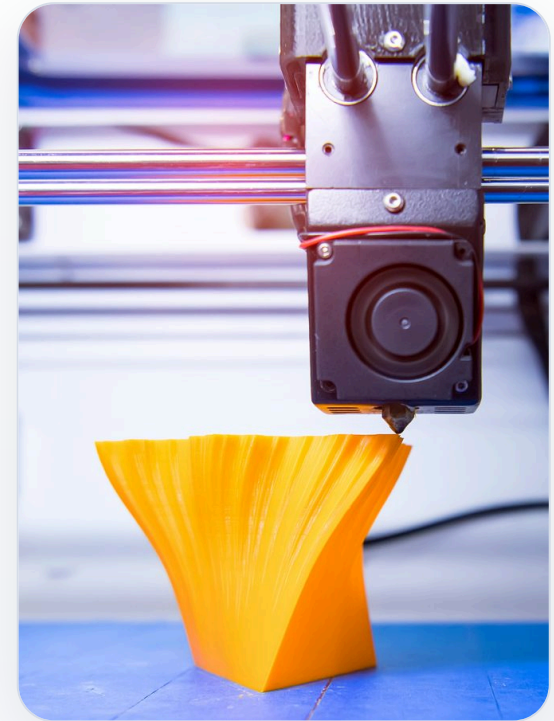
From model to machine part: design in Onshape, prepare in PrusaSlicer, and print the functional parts your robot needs.

Thu, Oct 1

JCC 301 / Nolop

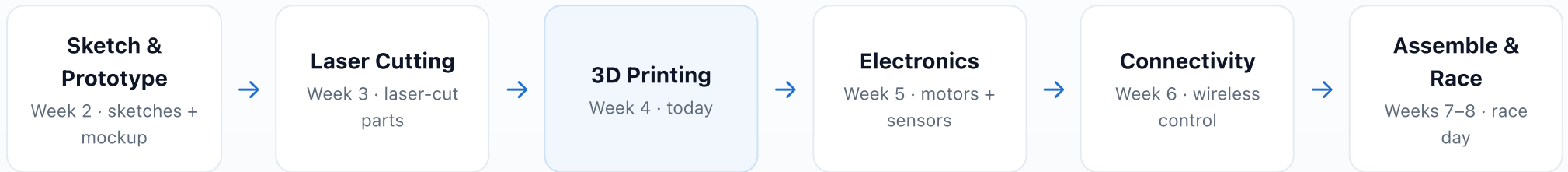
Design → Prepare → Print

Individual print due next week



# Six steps, one working robot

The path from sketch to race — today we're on step three.



**TODAY**    **Model 3D parts** — mounts, brackets, and custom components   ·   **3D print functional parts** for your robot and its dispenser.

# Today we print **the parts**

Your robot — plus a supporting dispenser — moves as many balls as possible between two lines. The custom pieces come off the printer.



## Form follows function

A basket, wheels, a screen, and a scanner all share one frame.

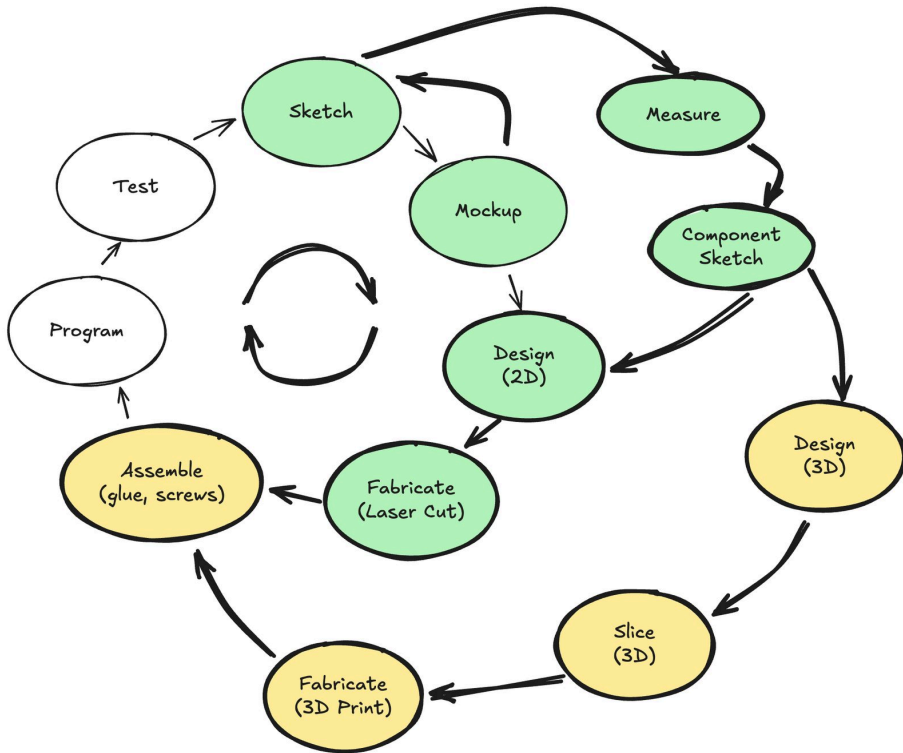


## Parts have jobs

A cart that scans items, shows the running total, and checks you out.

# Where 3D printing fits

Sketch, mock up, design, fabricate, assemble, program, test — then loop back around.



2D

## Laser-cut branch

Component sketch → 2D design → flat parts, cut and assembled.

3D

## Printed branch

Design (3D) → slice → fabricate: parts with height, curves, and internal detail.

→

## The loop closes

Assemble, program, and test — then measure and feed the next iteration.

# Inside a 3D printer

Four parts do the work — everything else supports them.



01

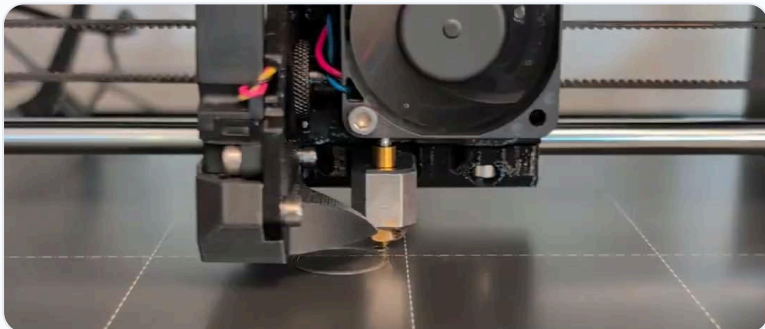
## Filament

The plastic feedstock, spooled above the printer and fed into the hot end.

02

## Nozzle

Melts the filament and lays it down, layer by layer — ours is 0.4 mm.



03

## Motors

Move the toolhead and the bed along the axes while the part builds.

04

## Heated bed

The build surface that the first layer sticks to.

## NOZZLE TEMPERATURE

**~220 °C / ~428 °F**

That's how hot the nozzle runs to melt filament. It is a burn risk — let the nozzle and heated bed cool before you reach in, and keep hands clear while a print is running.

# Lots of filament options

PLA is the profile our printers run — but the material you pick changes how a part behaves.

The screenshot shows the SIMPLIFY3D website's 'Ultimate 3D Printing Materials Guide'. The navigation bar includes links for SOFTWARE, INDUSTRIES, RESOURCES (highlighted), BLOG, COMPANY, and a BUY button. The page title is 'Ultimate 3D Printing Materials Guide'. Below the title, there is a paragraph of introductory text. A row of filter tags is displayed, including 'All', 'Chemically Resistant', 'Composite', 'Dissolvable', 'Elastic', 'Fatigue Resistant', 'Flexible', 'Heat Resistant', 'Heated Bed Not Required', 'Impact Resistant', 'Rigid', 'Soft', 'UV Resistant', and 'Water Resistant'. At the bottom, there are four images showing various 3D printed parts: colorful building blocks, a black ring, a small boat, and a mechanical part being tested in a liquid.

## BROWSE BY PROPERTY

Chemically resistant

Composite

Dissolvable

Elastic

Fatigue resistant

Flexible

Heat resistant

Impact resistant

Rigid

Soft

UV resistant

Water resistant

Materials guides compare dozens of filaments side by side before you commit a spool.

# Design, prepare, **print**

Three tools, three steps — the same loop for every part you make this term.

## STEP 1 · DESIGN

Onshape

### Model the part

- Sketch and extrude in the browser
- Export the part as an STL

## STEP 2 · PREPARE

PrusaSlicer

### Slice the model

- Import the STL and arrange it on the bed
- Slice it into G-code

## STEP 3 · PRINT

Prusa MK4

### Run the printer

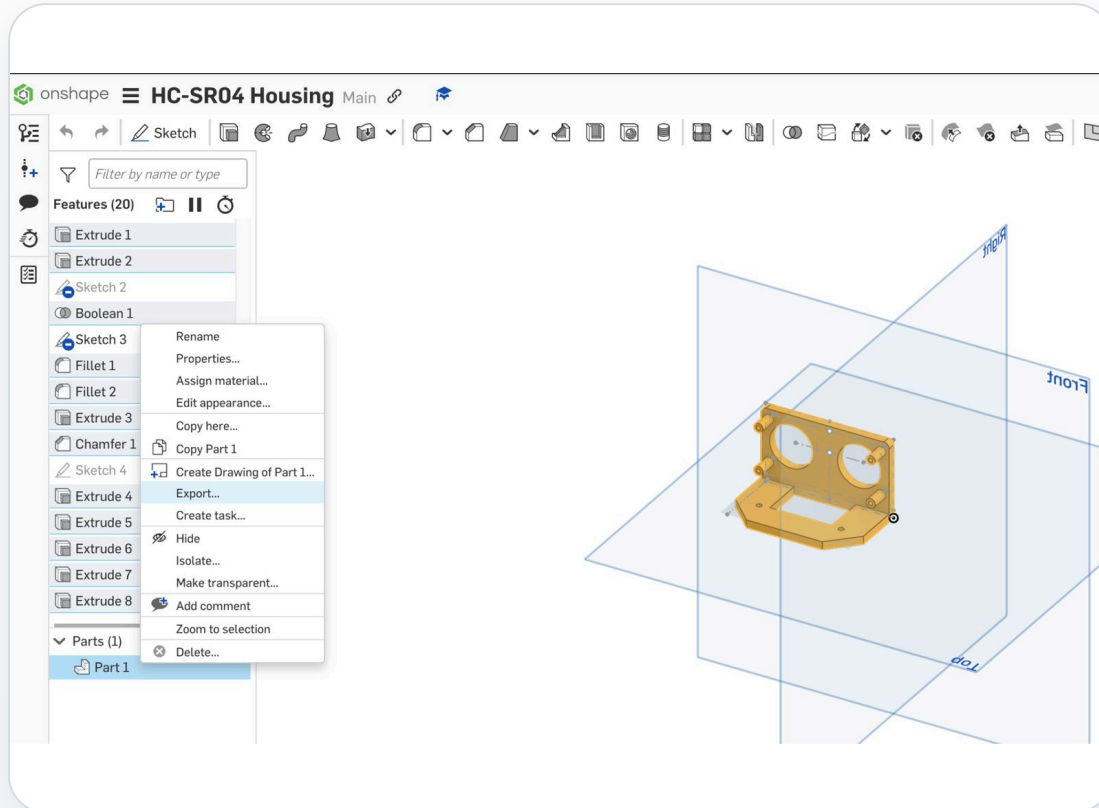
- Save the G-code to a USB drive
- Load and print at Nolop

**SIGN UP**   **Onshape** — [onshape.com/en/education/sign-up](https://onshape.com/en/education/sign-up) with your Tufts account · **PrusaSlicer** — [prusa3d.com/page/prusaslicer\\_424](https://prusa3d.com/page/prusaslicer_424)



# Model the part in Onshape

Build the part on screen from sketches and features, then export it as an STL for the slider.



01

## Sketch

Start with a 2D profile and dimension it.

02

## Build features

Extrude, boolean, fillet, and chamfer.

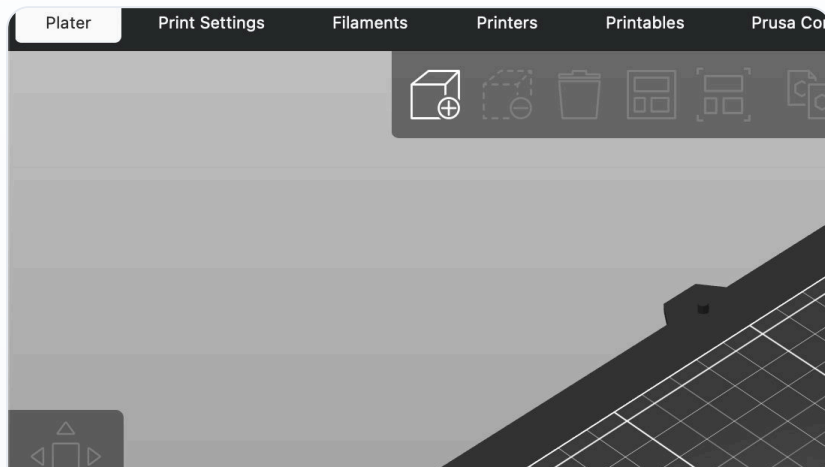
03

## Export

Export the finished part as an STL.

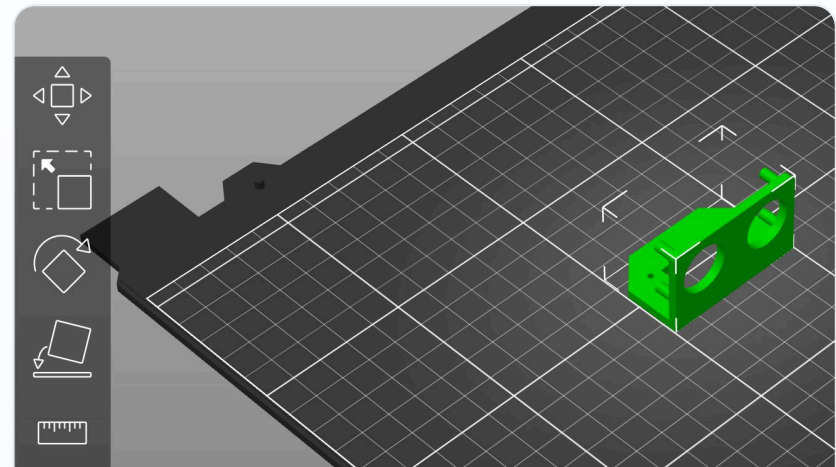
# Import and arrange

Drop the STL into PrusaSlicer, then move, scale, rotate, place on face, and add support before you slice.



## Import the STL

Open the file you exported from Onshape.



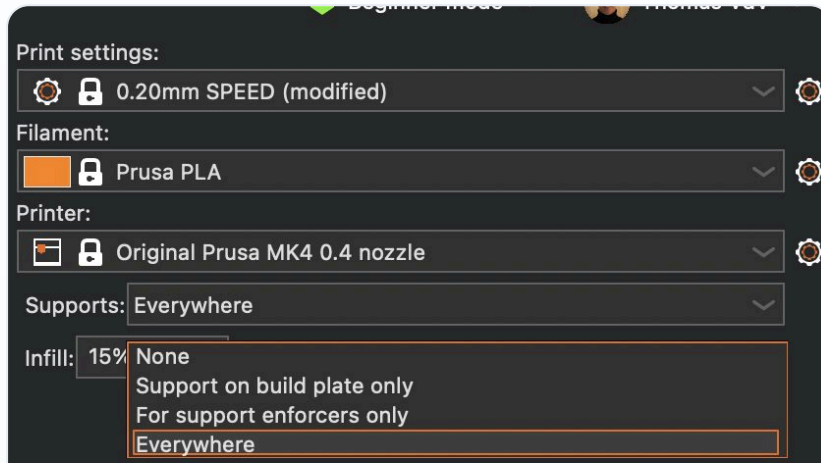
## Arrange it

Place the part on a face and add support where it overhangs.

PrusaSlicer — [prusa3d.com/page/prusaslicer\\_424](https://prusa3d.com/page/prusaslicer_424)

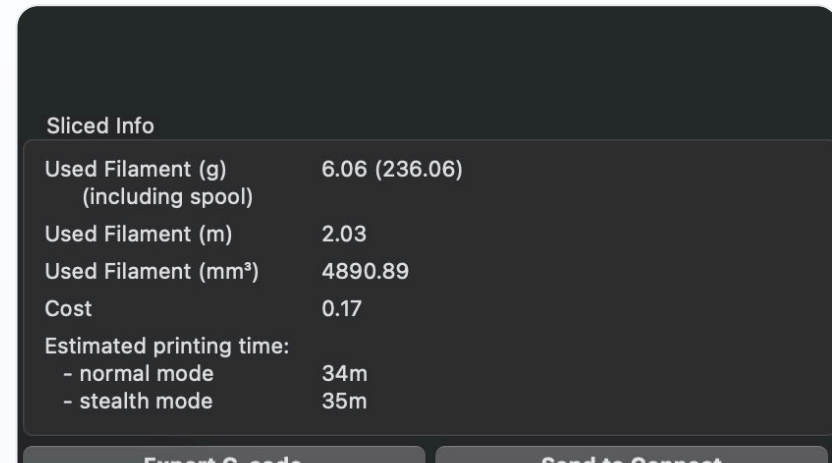
## Slice, check, export

The slicer tells you what the print will cost in time and filament before you commit — then save the G-code to a USB drive and print at Nolop.



### Check the profile

Print settings, filament, printer, supports, and infill.



### Slice and export

Read the filament and time estimate, then export the G-code.

# Print without **the drama**

Seven habits that keep the printers — and your prints — in one piece.

## WHILE YOU PRINT

### At the printer

- Do not leave your print unattended
- Check the filament so you won't run out
- Remove your print promptly when it finishes
- Leave a clean working bed behind

## IN YOUR DESIGN

### Support & alternatives

- Think about support placement during design
- Throw support material in the trash
- Consider laser cutting as a faster, cheaper option

# Print something **interesting**

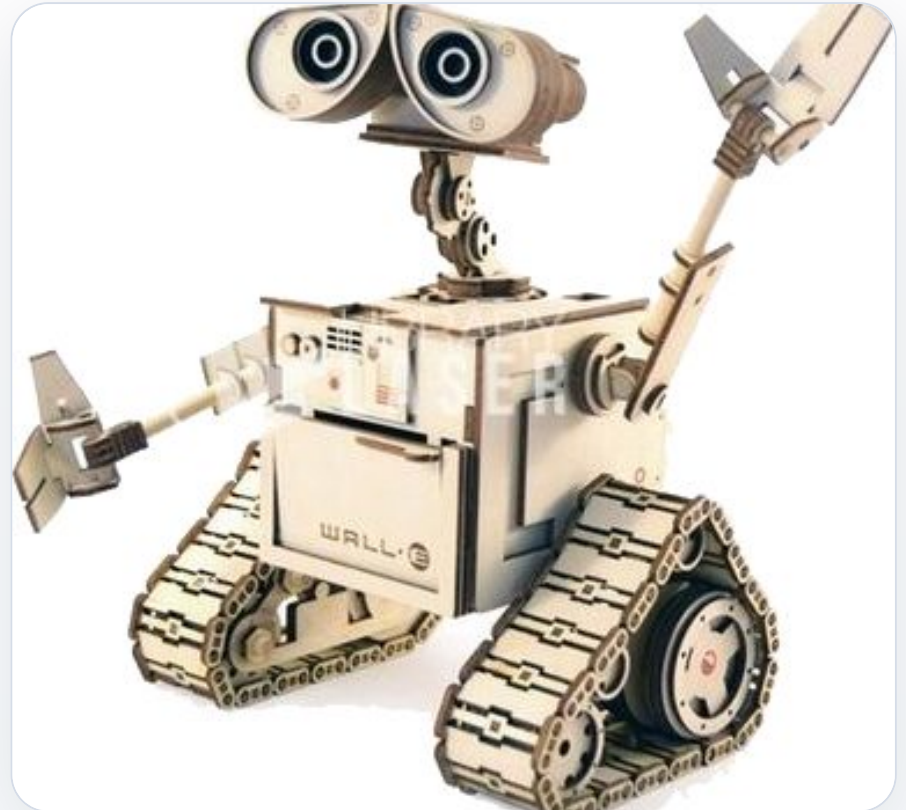
5 points · due next week

## THE BRIEF

3D print · 5 pt

### Design and fabricate an object

- Design and fabricate an interesting object using 3D printing
- Learn the digital-to-physical workflow
- Experiment with form, function, and creativity
- Bring the finished object to class for presentation and discussion



# Mid-point **assembly check**

A checkpoint that verifies your team has a physically assembled, structurally sound robot — and dispenser if applicable — with all electronics in place before you start writing software.

## **Assembled, not wired**

You don't need every component wired yet — but you should have a plan for routing the wires.

## **Progress, not perfection**

A solid frame, clean wiring, and a couple of known issues is exactly where you should be.

## **Catch it now**

The goal is to find mechanical and integration problems while there is still time to fix them.

A “finished” robot held together with tape and optimism is not the goal — show us where you are and your clear-eyed plan for where you're going.

# What to show, **what to submit**

## DEMONSTRATE AT NOLOP

### Bring the build

- Your fully assembled robot
- One supporting dispenser, if applicable

## SUBMIT IN CANVAS

### Photos + self-assessment

- At least 3 photos of the build from different angles
- Current state — what is physically complete and wired? Be specific
- Known issues — what problems remain unresolved?
- Plan forward — what will you prioritize in the remaining time?

# What we're **looking at**

Prioritize these — see the assignment for details.

**Structural integrity**

**Drivetrain & mobility**

**Power system**

**Sensor placement**

**Serviceability**

**Design intent**

**Electronics mounting**

**Wiring & connections**

**Wire management**

**PRIORITIZE THESE** See the assignment for the details behind each criterion.



# Don't build it **this way**

Five shortcuts that cost teams time on race day.

## Hot glue as a structural joint

It fails under stress. Use mechanical fasteners for load-bearing connections.

## Tape-mounted electronics

Masking tape releases mid-run. Use standoffs, screws, double-sided tape, or adhesive-backed Velcro.

## Wires through the drivetrain

A cable touching a spinning wheel will eventually get caught and ripped out.

## No slack in sensor wires

If a sensor moves, its wires need strain relief and enough length to handle the motion.

## An inaccessible battery

If swapping a battery requires partial disassembly, you'll waste valuable testing time later.

BEFORE YOU GO

# Questions?

Then go make something. Next week: microcontrollers — we wire motors and sensors, program movement, and you bring your 3D print to class.

## WHERE TO FIND US

### Office hours

[go.tufts.edu/meet\\_tvdv](https://go.tufts.edu/meet_tvdv)

### Email

[thomas.van\\_de\\_velde@tufts.edu](mailto:thomas.van_de_velde@tufts.edu)

### Course materials

[tvande08.github.io/ENT-164](https://tvande08.github.io/ENT-164)